e) Electricity due to Chemical Action

By using chemical action method, electrons are extracted from an atom and electricity is produced. The voltage developed in storage battery is due to this chemical action only.

f) Electricity due to Magnetism

In this method, electrons are extracted from an atom due to magnetism. In generators, conductors are moving within the magnetic field to generate electricity.

1.1.3 Power Generating Plants

Today, approximately seven electric power generating stations are available in our country.

TYPES OF POWER GENERATING PLANTS

a) Hydroelectric Power Plant
b) Thermal Power Plant
c) Atomic Power Plant
d) Gas Power Plant
e) Diesel Power Plant
f) Solar Power Plant
g) Wind Power Plant

a) Hydroelectric Power Plant

From the water reservoir, the water is taken through the joint tubes to the water turbine. For the rotation of turbine, the kinetic energy of water is converted into mechanical energy and is converted into electrical energy by using generator.

This type of plant is placed in Tamil Nadu at Mettur, Kunda, Pykara, Suruliyaru and Kadamparai.

b) Thermal Power Plant

Chemical energy is converted into heat energy by burning coal or lignite in boiler plant. Water in the boiler is converted into steam by heat energy. This steam is flowing through the steam turbine which is connected to the generator and this energy is converted into mechanical energy by the rotation of turbine. The mechanical energy is again converted into electrical energy by the use of generator.

This type of plant is placed in Tamil Nadu at Ennore (Chennai), Neyveli, Tuticorin and Mettur.

Thermal power plants play a major role for the requirement of electricity in Tamil Nadu.

c) Atomic Power Plant

By the diffusion of an atom of Uranium or Thorium, more heat is produced. The atomic power plant is working based on this principle. The heat energy produced is used to rotate the steam turbine and this energy is converted into mechanical energy. The generator converts the mechanical energy into electrical energy.

This plant is placed in Kalpakkam near Chennai,Koodangulam in Tirunelveli District and Tharapur in Rajasthan. Leakage of radiation by this plant may cause pollution and affect the health of the people.

d) Gas Power Plant

The process of generating electrical energy with the help of gas turbine (which acts as a prime motor) is known as Gas power

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